



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Scalable and Reliable Services

SRS

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The course

- **Classroom lessons:** 64 hours - **Language:** English

- **Main objective**

Modern digital services must satisfy various requirements in terms of **performance**, **scalability**, **reliability** and security. We analyze the whole lifecycle of high-quality services, from design to deployment, testing and operation. In each phase, we aim to provide participants with the ability to gather and analyze Big data, to identify sources of outages and critical dependencies, and to avoid bottlenecks and single points of failure. Finally, we expect participants to be prepared to evaluate and discuss alternative solutions and justify investment in support of business services

- **Prerequisites**

- Fundamental courses characterizing any Bachelor in Computer Science or Computer Engineering: systems, networking, software
- Knowledge about distributed systems, cloud, software engineering is preferred



Calendar

- 5 hours per week, every Monday and Wednesday, but:
 - March 22st: Graduation day
 - April 10th: Easter Monday
 - April 24th: Liberation period
 - May 1st: Workers day
- Monday - 9:15-11:30, Room 1.4
- Wednesday 11:00-13:15, Room 1.5

If everything goes well, teaching period ends May 31st

*Exact schedule is subject to variations independently of my will.
Please, check notes and advices on **virtuale.unibo.it***



Introduction

- Cloud services and cloud provider architectures
- High-quality services: performance, scalability, reliability (and security)
- Lifecycle of high-quality services

Service deployment

- Secure and robust software production
- Secure and robust DevOps
- Labs

Architectures for scalable and reliable services

- Sources of outages and critical dependencies
- Bottlenecks and single points of failure
- Design
- Maturity assessment
- Architectural principles for reliability and scalability
- Migration to cloud
- Critical data management
- Strategy, Governance, Management



Tests

- Benchmarking
- Performance testing
- Stress testing

Scalable and reliable organizations

- Roles and responsibilities
- Leaders and managers
- Putting all together
- Capacity planning
- Understanding and managing complexity
- Managing changes

Issues, incident and crises

- Performance monitoring
- Warning signs in operation
- Incidents management
- Crisis management



Slides and other material

<https://virtuale.unibo.it/>



Types of exam

Students that are oriented to take this course as a priority and to pass the exam **by the end of July 2023** have two options:

- A. Participation to a group software project based on *DevOps/DevSecOps* model + short report + individual presentation of the results**
- B. Analysis and discussion of a problem in classroom + Oral exam on the topics of the course**

From **September 2023 onwards**, there is only one possibility:

- B. Analysis and discussion of a problem in classroom + Oral exam on the topics of the course**

There are no biases in the final evaluation



Type A (1)

- The group software project can be chosen among a set of assignments that will be communicated by **March**
- The project is based on *DevOps/DevSecOps* model on a cloud platform
- The results (code and short report) must be sent 24 hours before the date of the exam
- Each member will present and discuss his/her activities in presence
- Anomaly management rules:
 - For serious motivations, some member(s) of the group can discuss the project work in another examination day, but the deadline for delivering the group assignment does not change
 - There might be motivations for leaving the group at any point. We will consider this case and facilitate the remaining people
 - There are no acceptable motivations for joining a group after the departure of the “train”



Type A (2)

- A typical group consists of 3 students with some flexibility
- It must operate in a synchronous way, hence there are three “trains” to jump on board:
 - Departure: April – Arrival (*delivery*): June 13 – Exam: **by June 15**
 - Departure: ??? – Arrival: July 3 – Exam: **by July 5th**
 - Departure: ??? – Arrival: July 20 – Exam: **by July 21st**
- Mandatory communications
 - Names of the group
 - Scheduled exam date

**It is possible to pass from exam type A to exam type B,
not vice versa**



Integration with other activities

- The SRS activities may be preparatory for the project work (4 CFUs) that is contemplated in some *Master Laurea* courses
- They are presented and evaluated at different times and with specific assessments
- The SRS activities + project work may be preparatory for theses



